

Container-based High-Performance Computing Approaches for Simulating Large-Scale Biological Neural Networks

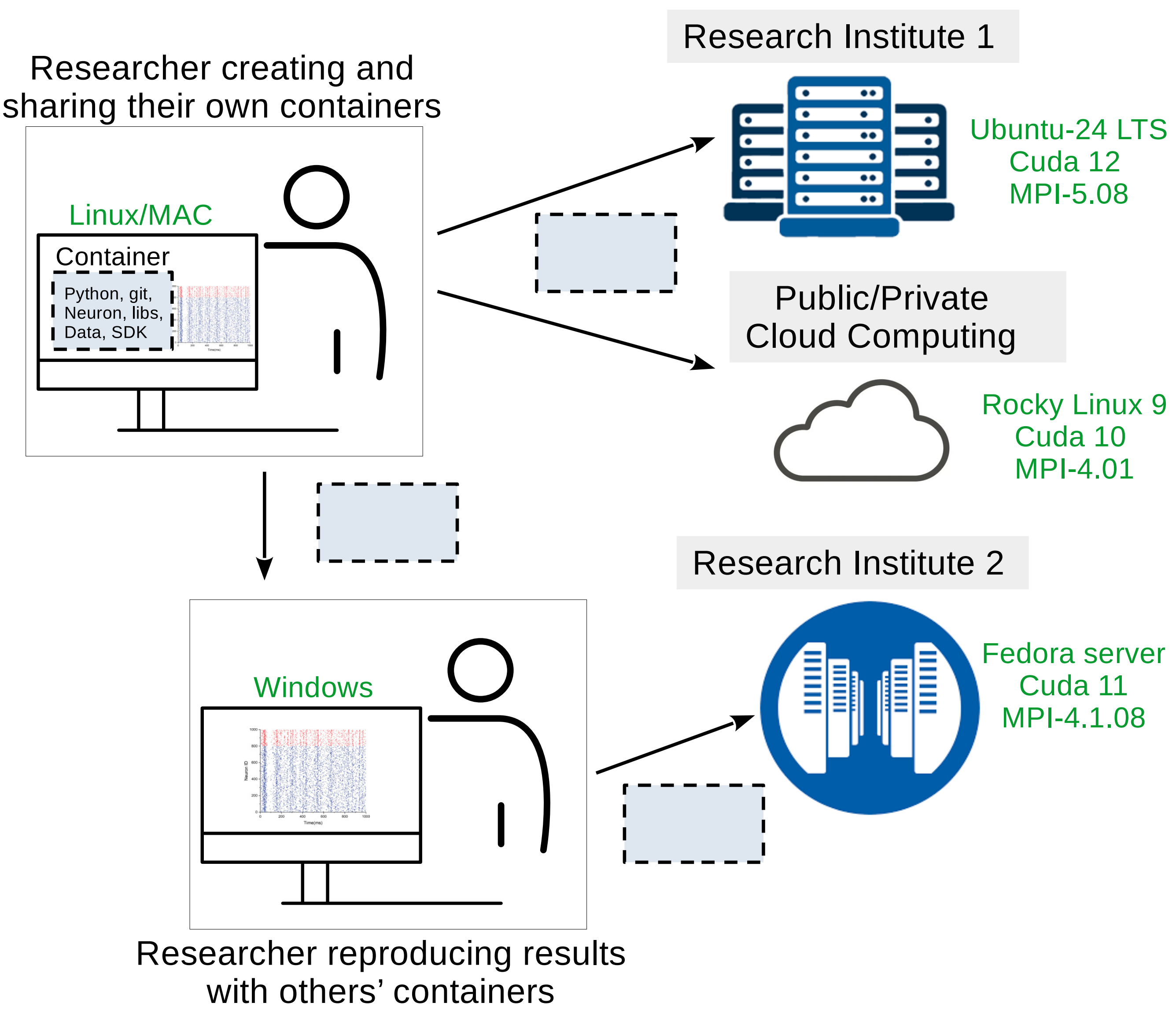
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Introduction

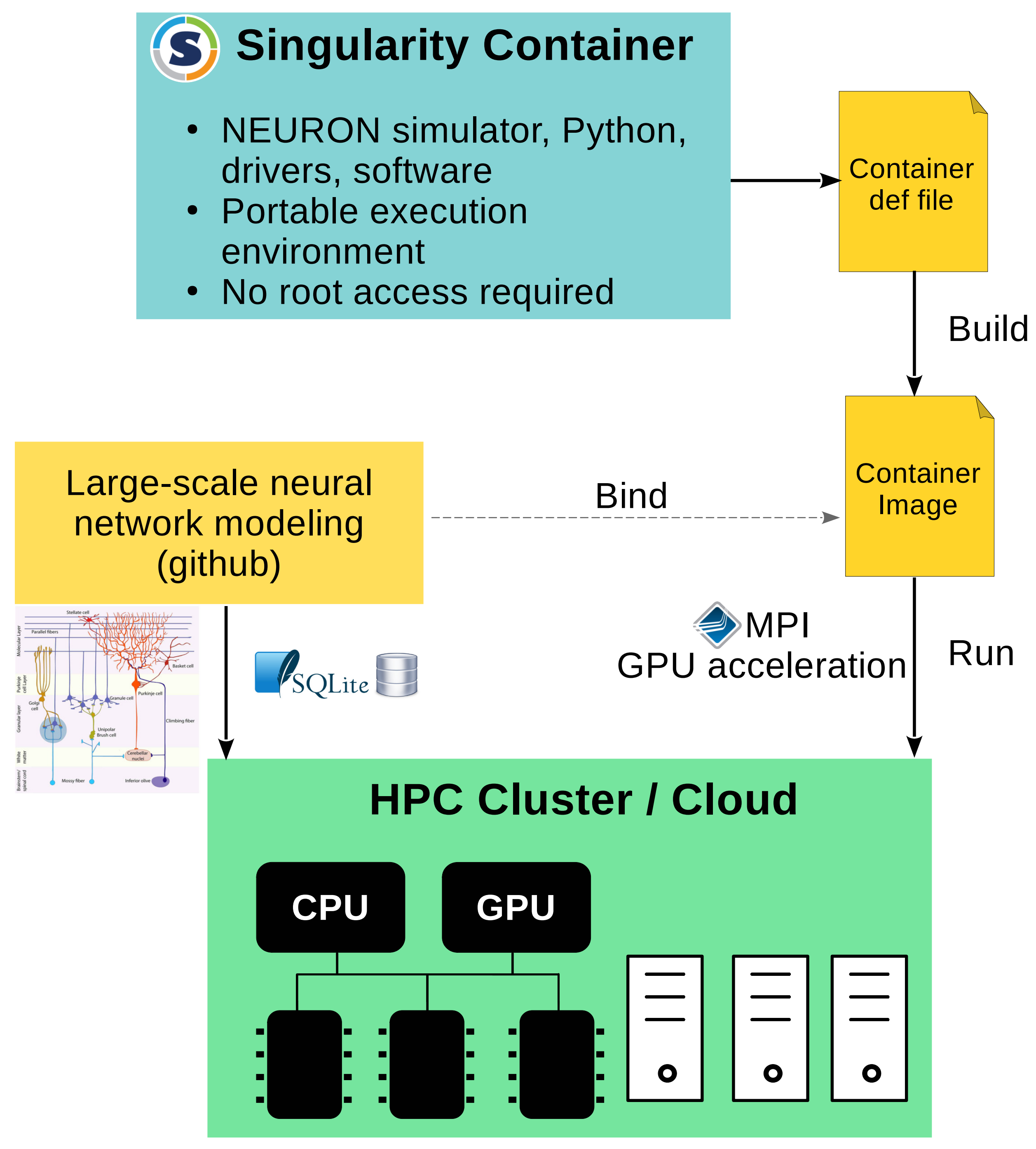
- 1. Modeling large-scale, biophysically detailed neural networks increasingly relies on high-performance computing (HPC).
- 2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to software incompatibilities and complex dependencies.
- 3. We address this issue using singularity containers that encapsulates the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC/Cloud systems.
- 4. We give a practical demonstrations by benchmarking HPC environments with cerebellar mossy fiber–granule cell network simulations.

Containerization for Scientific Computing

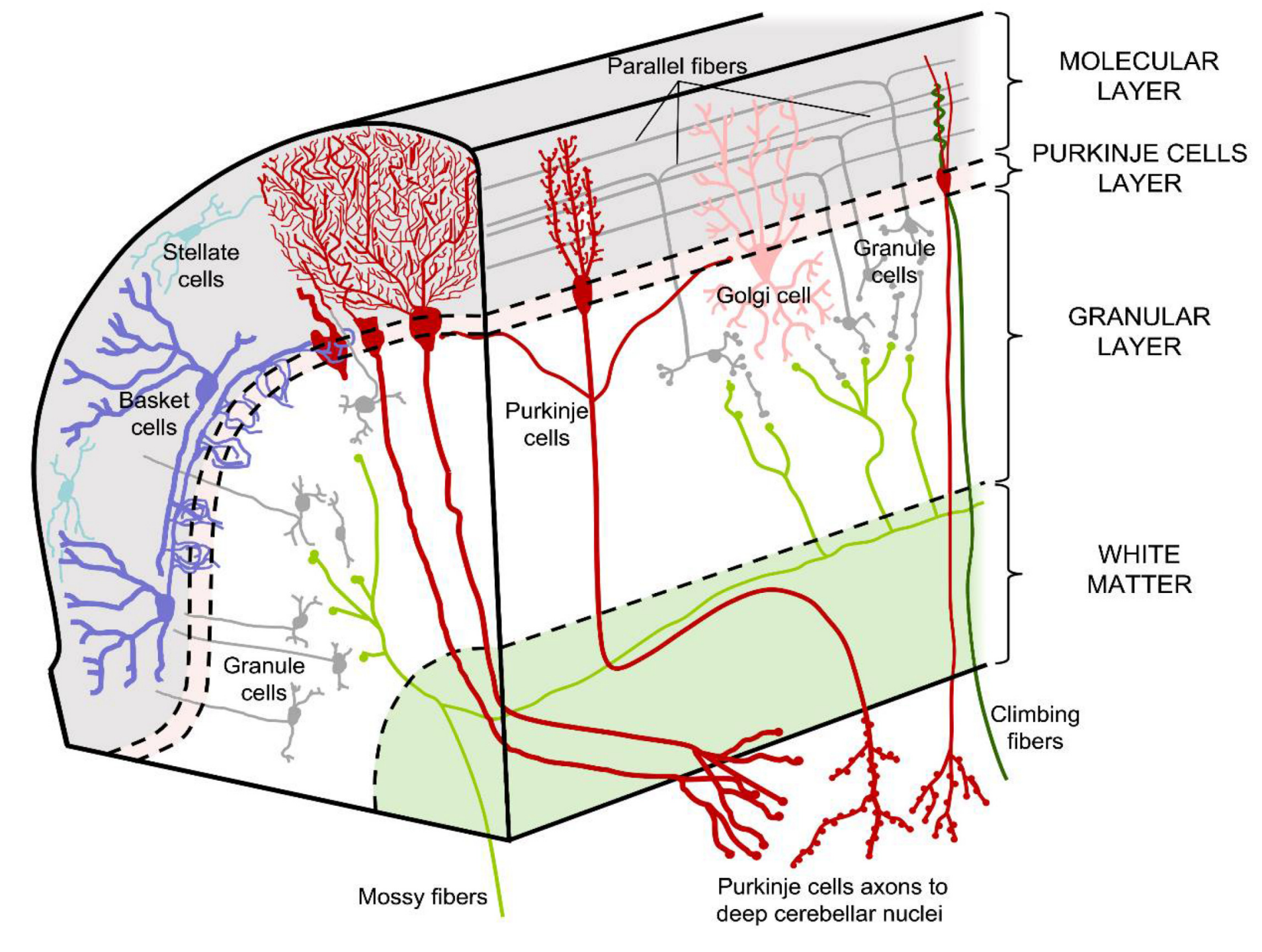
Container is a lightweight, standalone, and executable package. It is desginded to include everything needed to run a scientific software.



Our Computational Framework



Benchmarking Data and Network*: Cerebellar Mossy Fiber → Granule Cell



* Nguyen, T.M., Thomas, L.A., Rhoades, J.L. et al. Structured cerebellar connectivity supports resilient pattern separation. Nature 613, 543–549 (2023)

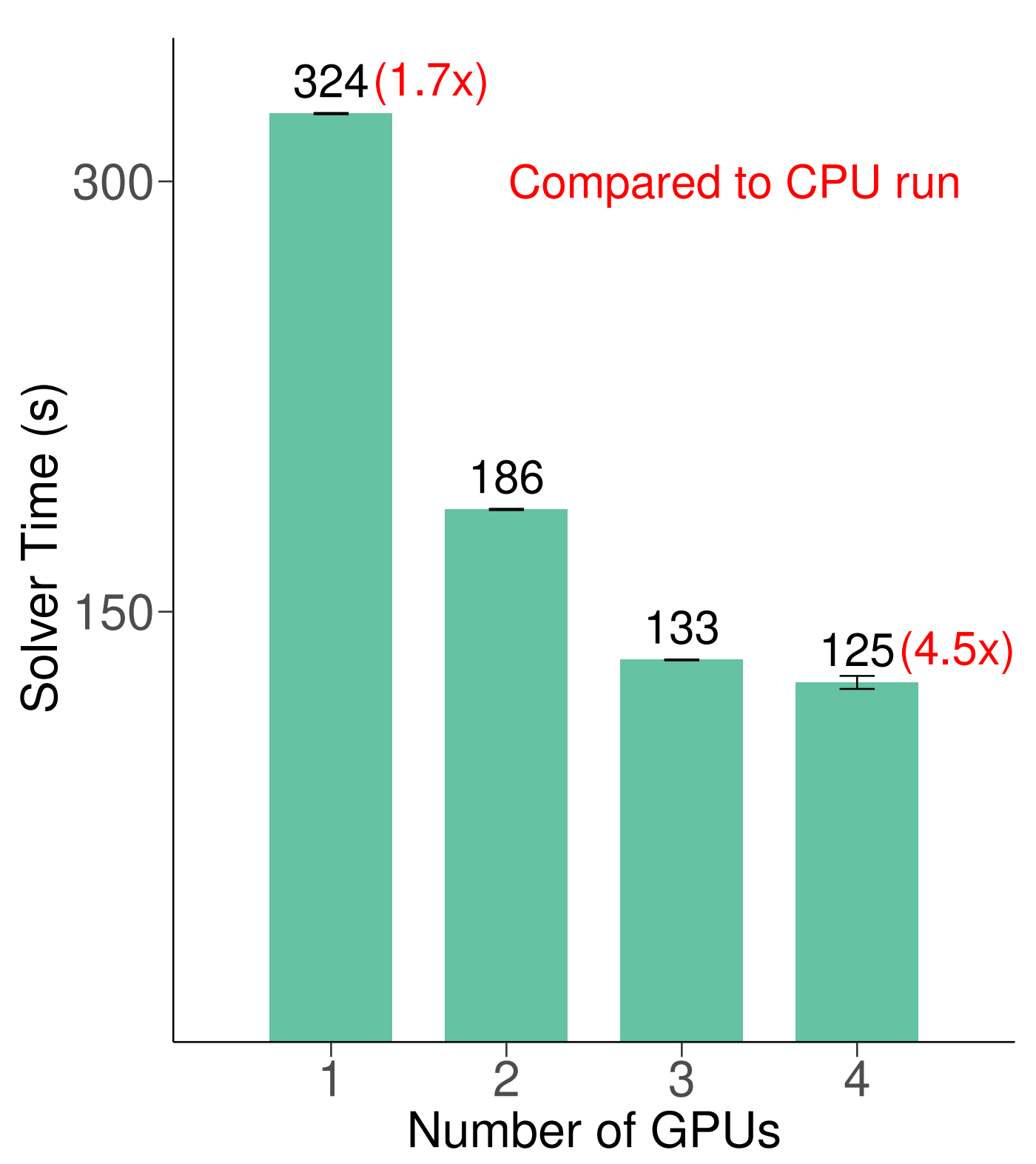
Table: Network Configuration and Simulation Parameters

Number of MF	85,600
Number of GRC	314,000
Number of compartments	942,000
Number of synapses	990,960
Simulation time	2 s

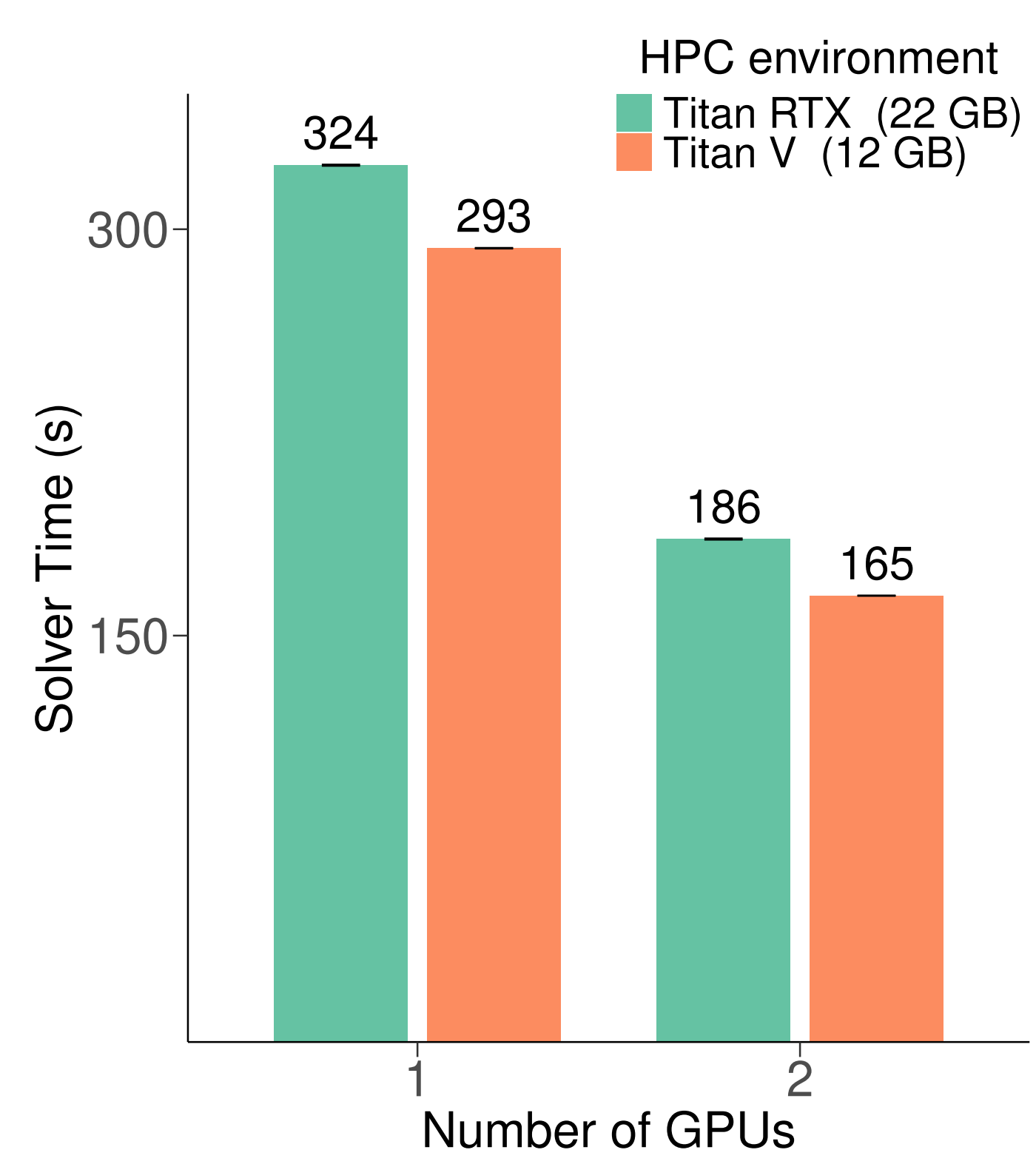
Benchmarking HPC Environments

Table: Performance in CPU parallel computing (30-task MPI)

CPU	Core	Solver Time (s)	Simulator
Intel(R)-Xeon	30	564	CORENeuron-CPU



(a) TITAN RTX architecture: Comparison of solver time for varying numbers of GPUs in CORENeuron-GPU simulator



(b) Comparison of solver time between HPC environments

Figure: Solver time comparisons in CORENeuron simulations across HPC platforms.

TITAN RTX outperform 30-core CPUs by up to 4.5x, with performance improving as GPU count increases.

Future works:

- 👉 To include more realistic model of granule layer, including Golgi cells.
- 👉 To extend the model to include Purkinje cells and other interneurons for a complete cerebellar circuit.

Acknowledgment

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